

Companion Evidence for Exact Bidirectional Steganographic Transport

Keywords: Artifact Recovery, Capacity, Detectability, Reproducibility.

Abstract: This companion supplies a transport diagram, expanded numerical tables, development serialization comparisons, arithmetic diagnostics and reproducibility details for the main paper. It adds no experiment, model inference, hypothesis test or independent sample. Prospective held-out results, development qualification, repeated development GPU timings and repeated context scores are kept separate. The main paper contains the protocol, principal outcomes and central limitations and does not depend on this companion being reviewed. A separate supplementary upload for an ICAART regular paper has not been verified in the official instructions. This document is therefore prepared as a companion artifact, not represented as accepted supplementary submission material.

S1 TRANSPORT AND INDEPENDENT VERIFICATION

The actual bidirectional examples appear in Figure 1 of the main paper. Figure S1 summarizes their shared transport and verification contract. A source-image PNG displays the canonical payload, but equality is evaluated on all 256 recovered grayscale bytes. Text equality concerns the complete literal UTF-8 bytes. Visual similarity or authentication alone does not substitute for these comparisons. Drafting assistance throughout this companion is disclosed at its end (OpenAI, 2025).

S2 COMPLETE PROSPECTIVE TABLES

Table S1 preserves the six method/direction cells, rate intervals and original prospective timing. All 20 arithmetic-text failures remain in attempted counts and contribute zero exact goodput. Table S2 provides both frozen detector scores, including below-chance point estimates and failed delivered carriers. No observer orientation is reversed. The grouped intervals retain payload class or text-length strata and shared controls, as specified in the main paper.

The exact-goodput difference between gated and fixed text is -0.2000 bits/token $[-0.2192, -0.1805]$. The corresponding charged-time difference is 23.96 seconds $[20.34, 27.58]$. For PNG, gated minus fixed charged time is 0.35 seconds $[-0.21, 0.98]$; arithmetic minus fixed is 0.83 seconds $[0.38,$

1.25]. All three PNG goodputs are identical by this allocation's fixed carrier size and exact recovery. The paired arithmetic-minus-fixed PNG mean-surprisal difference is -0.4216 bits/channel $[-0.5470, -0.3008]$. These are retained accepted comparisons, not post hoc selections of tests. The full machine-readable paired table is carried in `data/accepted_paired_differences.json`.

For equal all-success binary cells, the observed difference is zero but the conservative paired recovery bound remains ± 19.68 percentage points. It derives from the frozen 97.5% marginal Clopper-Pearson bounds. This prevents the zero-width paired bootstrap display from being interpreted as a population equivalence statement. The sample remains 20 allocated groups per direction.

S3 PACKET AND CAPACITY ACCOUNTING

The packet consists of a 12-byte nonce, 264 encrypted bytes and a 16-byte tag. Its eight-byte protected header records version, kind, length and dimensions. The 256-byte slot is full for images, but has a mean 162.8 padding bytes for held-out texts. Thus 36 framing/encryption bytes and slot padding are separate costs. Fixed radix 16 requires no rank-alignment padding. The accompanying protocol identity record retains the exact framing identifier and unrounded entropy thresholds.

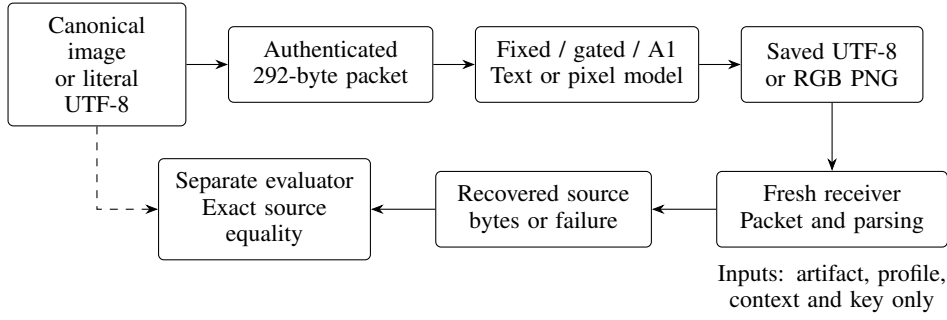


Figure S1: Common transport and independent verification. The same packet contract feeds a text language model or a pixel autoregressive model. Only saved UTF-8 or PNG crosses the carrier boundary. A fresh receiver uses its declared model/profile, shared context and key. Source bytes and sender diagnostics remain outside its inputs. A separate evaluator checks source equality after receiver exit. The conditioning row is prepended internally, not delivered as an extra PNG row.

Table S1: Expanded prospective recovery, goodput and runtime. There are 20 payload groups in each cell. Recovery intervals use two-sided 95% Clopper–Pearson bounds. Goodput intervals use 2,000 stratified payload-group bootstrap draws. Phase times and charged pair times are overlapping boundaries, not costs to add. Text goodput is bits/token and PNG goodput bits/channel value.

Carrier / method	Exact / attempted; 95% interval	Exact goodput; 95% interval	Encode phase (s)	Decode phase (s)	Charged pair (s)	Failure
UTF-8 / Fixed	20/20 [83.16, 100.00]%	3.3247 [3.3247, 3.3247]	45.63	45.31	123.33	None
UTF-8 / Gated	20/20 [83.16, 100.00]%	3.1246 [3.1054, 3.1441]	57.31	57.30	147.29	None
UTF-8 / A1	0/20 [0.00, 16.84]%	0.0000 [0.0000, 0.0000]	398.15	396.01	826.57	Token-cap capacity
PNG / Fixed	20/20 [83.16, 100.00]%	0.2505 [0.2473, 0.2535]	85.27	85.31	178.18	None
PNG / Gated	20/20 [83.16, 100.00]%	0.2505 [0.2473, 0.2535]	85.29	85.65	178.53	None
PNG / A1	20/20 [83.16, 100.00]%	0.2505 [0.2473, 0.2535]	85.59	85.77	179.01	None

S3.1 A1 Interval Rules and Diagnostic Limits

The A1 implementation uses 32-bit half-open intervals. After selecting a subinterval $[l, u)$, its common-prefix length is

$$b = 32 - \text{bitlength}(l \text{ xor } (u - 1)). \quad (1)$$

It emits the first b bits of l . With $M = 2^{32} - 1$, the next interval is

$$l' = (l \ll b) \& M, \quad (2)$$

$$u' = (((u - 1) \ll b) \& M) + 2^b, \quad (3)$$

where the mask applies before the final addition in the expression for u' . The main paper gives the exact partitioning and zero-extension rules. Unlike a general arithmetic coder with an additional underflow-normalization mechanism, this adaptation does not add an E3 rule or a final endpoint dump. The inspected Ziegler reference (Ziegler et al., 2019) is algo-

rithmic attribution; its source was not copied because a reuse license was not identified.

A concrete independently checked limitation is a narrow interval straddling a leading-bit boundary whose quantized distribution retains only one nonzero bin. Selecting that bin can leave the interval unchanged indefinitely. This is a property of the frozen adaptation, not proof that every zero-bit step stag-nates. In the actual prospective text cases, final recovered information closely tracks accumulated eligible surprisal. Two cases have one and three isolated unchanged steps, with no consecutive run beyond one. A successful PNG also has one isolated step. The 40 retained prospective audits agree with the independent partition/interval oracle and source-prefix checks. No arithmetic text case reaches finite-message zero-extension or termination, so its failure is not evidence of a termination bug.

Detailed packet windows and rank traces remain private. The retained public endpoint and 128-position checkpoint summaries are sufficient to in-

Known-model / known-context observer · frozen higher-is-stego score

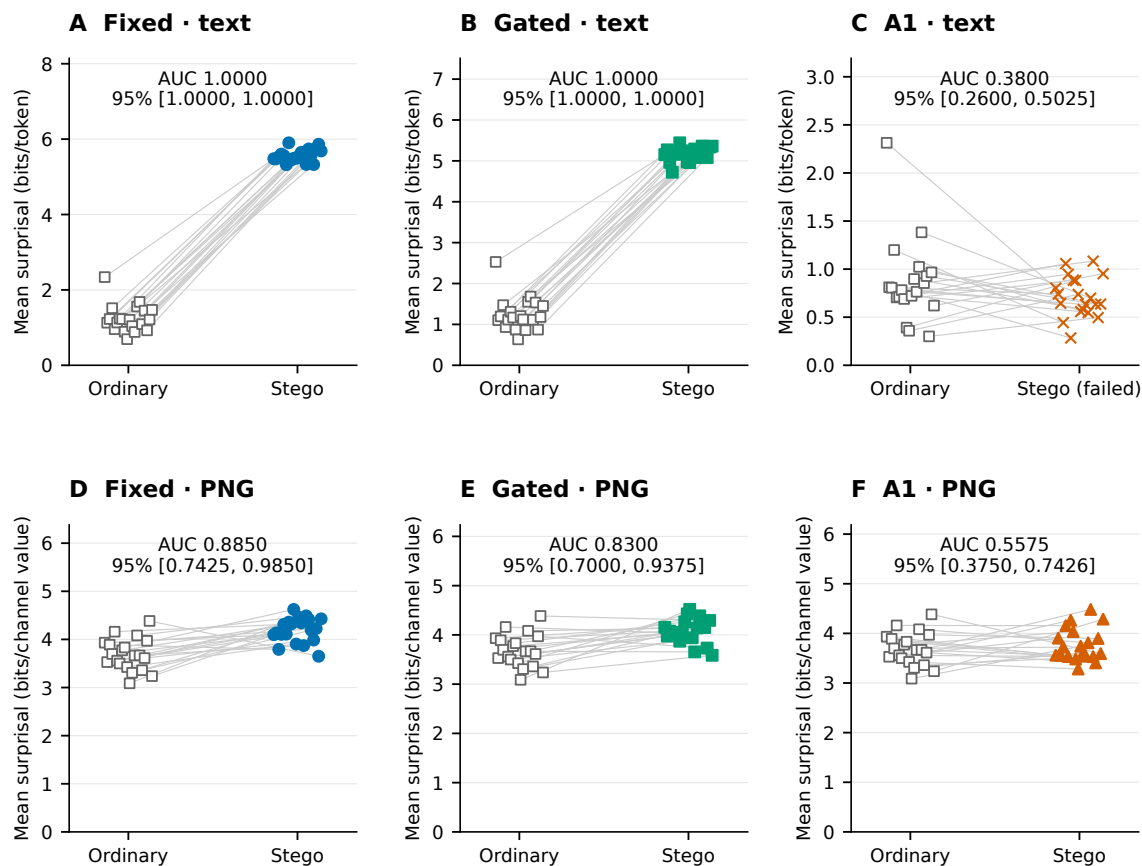


Figure S2: Whole-carrier mean-surprisal scores for the known-model, correct-context observer. Each panel shows 20 stego carriers and 20 matched ordinary observations with payload-group links. Failed A1 text carriers remain included. Text controls share prefixes of one trace per group, while PNG methods share full controls. Panel scales differ. AUC orientation is always higher-is-stego, with accepted 95% grouped intervals. Complete empirical separation in the small fixed/gated text samples, including collapsed bootstrap intervals, does not establish perfect population detection.

spect capacity without exposing those private audit streams. Complete packet-prefix bits are capped at 2,336 for successful PNGs, with suffix diagnostics reported separately. No line between observed checkpoints is used to infer a missing trace value.

S4 DEVELOPMENT SERIALIZATION EVIDENCE

The second prompt is “Explain how a home cook prepares a simple vegetable soup. Use continuous prose.” It was frozen before its carriers were generated. The singleton arm lazily caches eligibility for encountered candidates but does not use the complete-prefix assertion. A final mismatch therefore remains a saved arti-

fact and experimental outcome. The main sequence arm retains strict final consistency. This comparison adapts an existing singleton mask and published stepwise verification (Mantzaris, 2026; Yan and Murawaki, 2025); it does not introduce a new tokenization theory.

Development qualification accounts for 152 stego outcomes. Fixed sequence text and fixed PNG each recover 40/40. Gated coding recovers 8/8 in each direction. A1 PNG recovers 8/8, while A1 text retains eight capacity failures. Static text recovers 22/40. Sixteen ordinary traces and 48 controls are accounted for. One historical static matched-control prefix was unavailable and remains unscored, not silently regenerated. These experiments establish configuration qualification and serialization behavior, not an ex-

Table S2: Full exploratory correct-context detection. Scores are whole-carrier mean surprisal and mean $\log_2$ rank, with higher always indicating stego. Intervals are the accepted 95% grouped percentiles. Each row has 20 scorable stego files and 20 matched ordinary artifacts. These associations use only 40 independent ordinary traces across the two directions, not 120 controls or 240 score observations. All 20 failed arithmetic text carriers are included. No duplicates or unscorable artifacts were dropped. A collapsed empirical interval does not establish perfect population detection.

Carrier / method	Frozen score	AUC [grouped 95% interval]	Scorable stego	Unique matched controls	Failed stego included
UTF-8 / Fixed	Mean surprisal	1.0000 [1.0000, 1.0000]	20	20	0
UTF-8 / Fixed	Mean log2 rank	1.0000 [1.0000, 1.0000]	20	20	0
UTF-8 / Gated	Mean surprisal	1.0000 [1.0000, 1.0000]	20	20	0
UTF-8 / Gated	Mean log2 rank	1.0000 [1.0000, 1.0000]	20	20	0
UTF-8 / A1	Mean surprisal	0.3800 [0.2600, 0.5025]	20	20	20
UTF-8 / A1	Mean log2 rank	0.3600 [0.2425, 0.4801]	20	20	20
PNG / Fixed	Mean surprisal	0.8850 [0.7425, 0.9850]	20	20	0
PNG / Fixed	Mean log2 rank	0.8400 [0.6825, 0.9650]	20	20	0
PNG / Gated	Mean surprisal	0.8300 [0.7000, 0.9375]	20	20	0
PNG / Gated	Mean log2 rank	0.7700 [0.6299, 0.8975]	20	20	0
PNG / A1	Mean surprisal	0.5575 [0.3750, 0.7426]	20	20	0
PNG / A1	Mean log2 rank	0.5750 [0.4050, 0.7576]	20	20	0

All V2 arithmetic cases · observed checkpoints every 128 positions + endpoint

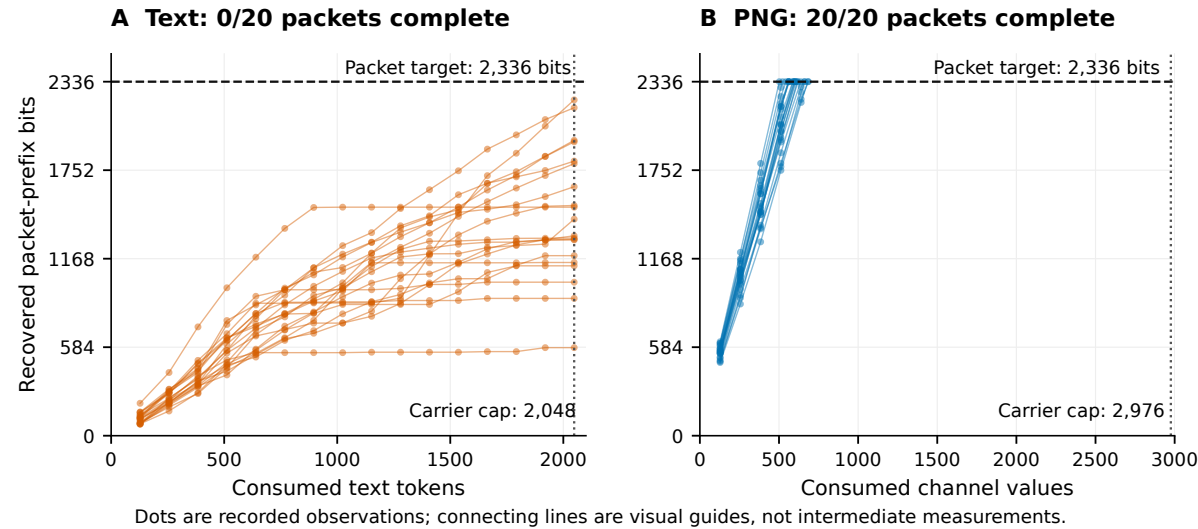


Figure S3: Arithmetic packet-prefix accumulation for all 20 prospective text and 20 PNG cases. Dots are retained observations every 128 consumed symbols and at the final endpoint. Connecting lines are visual guides, not unrecorded intermediate measurements. The target is 2,336 useful packet bits. Text reaches its 2,048-token cap with only 581–2,216 bits. PNG reaches the target before ordinary completion fills the remaining canvas. Curves stop at the packet endpoint and do not include emitted zero suffix bits as useful information. Carrier units differ between panels. Isolated unchanged intervals in two failed texts and one successful PNG do not establish sustained stagnation as the text-failure cause.

panded held-out sample.

Twenty predetermined fixed PNG development carriers, covering both shared rows, were re-saved

with another lossless compression setting and no ancillary metadata. Dimensions, RGB mode and every pixel byte remained equal. Fresh receivers then re-

Table S3: Overhead means for every prospective method/direction cell. The packet’s 36 framing bytes and slot padding are separate. A1 packet-positive and zero-bit positions partition its packet span. Its termination position overlaps the last positive position and is not added again. Emitted zero suffix and lookahead-zero diagnostics also overlap conceptually and are not extra payload. Text A1 has no complete-packet transport rate. The fixed full PNG has 2,976 channels. Serialized expansion divides delivered file bytes by canonical source bytes.

Carrier / method	Slot pad (B)	Packet-positive positions	Skipped positions	Zero-bit positions	Completion positions	Packet bits / span symbol	Suffix bits
UTF-8 / Fixed	0.0	584.00	0.00	0.00	32.00	4.000	0.0
UTF-8 / Gated	0.0	584.00	39.80	0.00	32.00	3.747	0.0
UTF-8 / A1	0.0	399.40	0.00	1648.60	0.00	—	0.0
PNG / Fixed	162.8	584.00	0.00	0.00	2392.00	4.000	0.0
PNG / Gated	162.8	584.00	100.50	0.00	2291.50	3.451	0.0
PNG / A1	162.8	508.20	0.00	89.65	2378.15	3.934	28.9

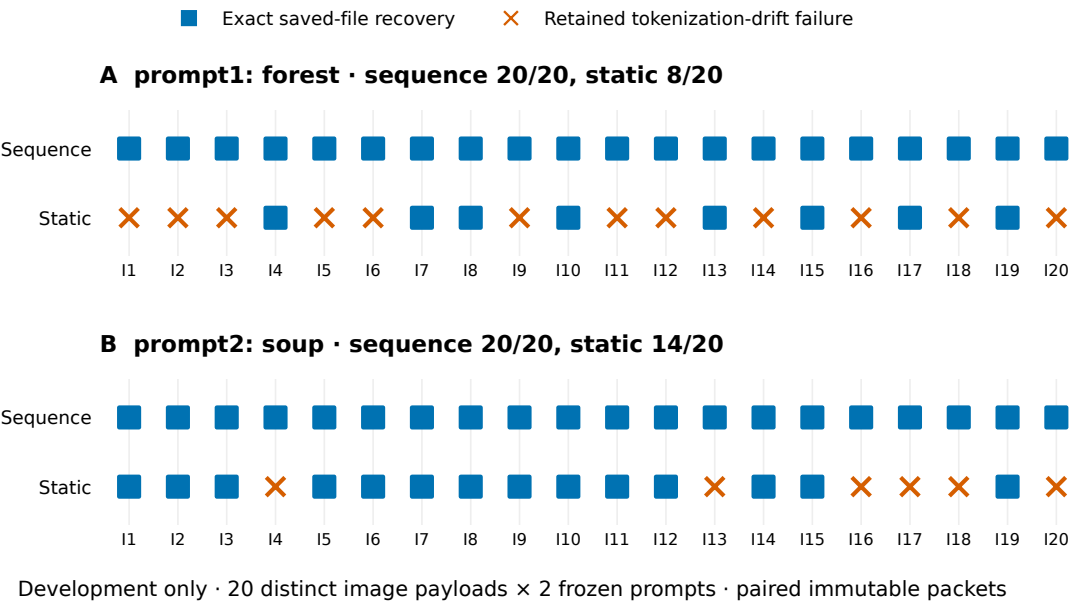


Figure S4: Paired development comparison of saved-text consistency, separate from the prospective study. Each column is one of 20 distinct image payloads under both frozen prompts. Sequence filtering checks the complete carrier prefix; static filtering checks singleton tokens only. Other candidate, probability, coding and completion rules are held fixed. Each context-specific pair uses the same sealed packet. Sequence recovery is 20/20 under each prompt. Static recovery is 8/20 for the forest prompt and 14/20 for the soup prompt, with 18 retained drift/replay failures. Static text is neither repaired nor accompanied by sender identifiers. The 40 context-specific cases are not 40 independent payloads.

covered all 20 original payloads exactly. These are preservation checks of existing carriers, not independent new transmissions. Text files also passed raw UTF-8 byte save/load checks. Ordinary CPU verification supports packet bounds, wrong-key authentication failure, nibble inversion, UTF-8 consistency, conditional RGB probabilities and separately written arithmetic oracles. Routine test counts remain in the supporting evidence rather than scientific outcome tables.

S5 CONTEXT SCORES AND DESCRIPTIVE SHIFTS

Table S4 expands the main context figure with both scores and paired intervals. The same 80 PNGs are rescored, and each of 20 controls is shared across methods. Twenty length-stratified payload groups, not 60 independent pairs, are resampled jointly with both observer conditions. The fixed primary decrease is supported by an interval below zero. Gated and A1

changes remain uncertain. A resolved change in one cell and an unresolved change in another do not establish a between-method interaction.

The paired fixed-minus-control mean score gap decreases from 0.507624 to 0.231119, a change of -0.276505 bits per channel. Controls have a larger mean score increase than fixed carriers, consistent with this narrower gap. The pattern does not establish an independently confirmed mechanism for the AUC difference. It cannot separate context correctness from differences between generated-image row statistics and random RGB statistics. Near-chance AUC also does not establish cryptographic secrecy or broad resistance to detection.

S6 COMPLETE GPU BENCHMARK

The three payloads were selected by first numeric development identifier within the 32–64, 65–96 and 97–128-byte bands, giving 32, 96 and 128 bytes. The same retained packet is used in each matched reference/graph comparison. Figure 4 of the main paper shows all nine fixed timing repetitions. Table S6 adds the two single-payload methods. No parameter, context or seed sweep was performed.

Reference execution remains available and is the prospective baseline. The optional CUDA graph path uses static CUDA input/output buffers, three warmup forwards, capture of the unchanged network and synchronized replay. CPU normalization, mixture probabilities, rank ordering and serialization are identical. Exact stream digests include length-framed identifiers, float64 probabilities and rank order at every delivered channel. Their equality is stronger than accepting a numerical tolerance, but it is still evidence on the checked cases and stack, not a cross-platform theorem.

For the nine fixed comparisons, charged pair time averages 177.819 seconds reference and 34.642 seconds graph. Inside those totals, encode plus decode phases average 170.184 and 25.829 seconds, CUDA-event intervals 165.431 and 21.251 seconds, cold model loading 1.876 and 1.884 seconds, and graph setup zero and 1.174 seconds. The remaining charged time is approximately 5.758 and 5.755 seconds. CUDA intervals overlap the phases, and no component is charged a second time.

Peak allocated PyTorch memory is 484.479 versus 473.849 MiB. Peak reserved memory is 516 versus 744 MiB. Sampled process memory is 912 versus 1,176 MiB. These different views overlap. Process-specific `nvidia-smi pmon` observations use the child

PID, with roughly 0.19-second polling. Driver windows, correlated samples and gaps prevent interpreting a single peak or sampled average as sustained utilization. All 45 benchmark model jobs, one short probe and 44 sender/receiver processes, have allocation and positive activity evidence. The probe is charged but not included in the matched timing means.

S7 REPRODUCIBILITY AND EVIDENCE BOUNDARIES

Checkpoint SHA-256 identities follow, wrapped without separators. Private paths, device serials and context bytes are omitted.

- Text GGUF SHA-256:
86c8ea6c8b755687d0b723176fcd0b24
11ef80533d23e2a5030f845d13ab2db7.
- PixelCNN++ checkpoint SHA-256:
a5ed558f6d4098ce3cbfd47658b7e3be
37fae77b71a90c8a56f46a6266ff833f.

Text retains eight threads, a 4,096-token context, batch/microbatch one, full eligible-layer offload, all-position logits and context/KV reset between sequences. The image stack is Python 3.10.13, PyTorch 2.5.1+cu124, CUDA 12.4, cuDNN 90100, NumPy 2.2.6 and Pillow 12.3.0. Both use float64 probability bookkeeping. Disabled TF32/benchmarking, deterministic algorithms, synchronous launches and `CUBLAS_WORKSPACE_CONFIG=:4096:8` are retained. Text additionally disables llama.cpp graphs/fusion and retains its recorded float32 cuBLAS control.

Focused RankCloak adaptations use revision ce853d4. The PixelCNN++ port is 7cb4436, with upstream bbc1568. Their notices are preserved. The checkpoint is `pcnn_lr.0.00040_nr-resnet5_nr-filters160_889.pth`, from the port author’s linked distribution. Strict loading removes a uniform DataParallel prefix bijectively. No replacement model was trained.

Public verification checks saved hashes, complete allocation, source equality, profile identities and shared controls. It verifies recorded authentication, not a new private-key operation. Fresh neural reproduction requires pinned models, compatible GPU runtimes and original context/key/packet bindings. Source-containing review directories are evaluator inputs, never receiver inputs.

The accompanying numerical CSVs and manuscript index map every result to retained evidence. The compilation-only source ZIP needs

Table S4: Correct and mismatched PNG AUCs for both frozen scores. Each row contains 20 scorable stego PNGs and 20 unique matched controls. Values and paired 95% intervals are the accepted 2,000-draw length-stratified group analysis, with conditions, methods and shared controls kept together. Higher score always indicates stego. The paired interval is computed from within-draw differences, not by subtracting marginal interval endpoints. There are no failed scores or exclusions.

Method	Score	Row1 AUC [95% CI]	Row2 AUC [95% CI]	Change [95% CI]
Fixed	Surprisal	0.8850 [0.7425, 0.9850]	0.6975 [0.5150, 0.8700]	-0.1875 [-0.2775, -0.1000]
Fixed	$\log_2$ rank	0.8400 [0.6825, 0.9650]	0.6625 [0.4725, 0.8475]	-0.1775 [-0.2800, -0.0950]
Gated	Surprisal	0.8300 [0.7000, 0.9375]	0.7475 [0.6074, 0.8825]	-0.0825 [-0.1625, 0.0100]
Gated	$\log_2$ rank	0.7700 [0.6299, 0.8975]	0.7225 [0.5775, 0.8675]	-0.0475 [-0.1301, 0.0375]
A1	Surprisal	0.5575 [0.3750, 0.7426]	0.5925 [0.4124, 0.7725]	+0.0350 [-0.0225, 0.1050]
A1	$\log_2$ rank	0.5750 [0.4050, 0.7576]	0.5975 [0.4175, 0.7726]	+0.0225 [-0.0325, 0.0725]

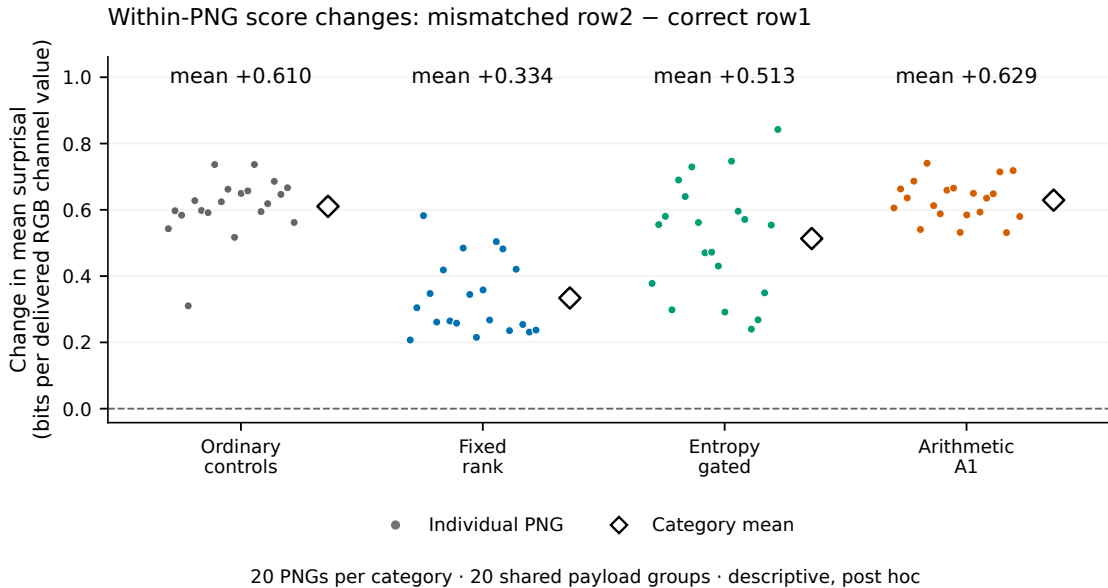


Figure S5: Descriptive within-PNG score shifts, row2 minus row1, in bits per delivered RGB channel value, not changes in AUC. All 80 observations come from the retained context study, 20 per category, with the same 20 payload groups and shared controls. Deterministic horizontal offsets only separate points. Open diamonds and annotations are category means. Row2 is the previously frozen random-RGB row, whereas row1 comes from a generated image. This post hoc analysis adds no test, interval or transmission. Mean shifts show score movement, while AUC depends on the full rankings. The plot does not establish secrecy or isolate a causal mechanism.

Table S5: Descriptive score means and within-artifact changes for Figure S5. Each category contains 20 observations. Units are bits per delivered RGB channel value. Controls are counted once, and the four categories share 20 groups. These are post hoc summaries without added inferential claims.

Category	Count	Row1 mean	Row2 mean	Mean change	Median change
Ordinary controls	20	3.683649	4.294085	+0.610436	+0.621412
Fixed rank	20	4.191273	4.525204	+0.333930	+0.285979
Entropy gated	20	4.074328	4.587583	+0.513255	+0.554706
Arithmetic A1	20	3.769649	4.398864	+0.629215	+0.635652

standard TeX packages but no experimental repository, private runs, models or network.

AI ASSISTANCE DISCLOSURE

OpenAI Codex assisted implementation, analysis/plotting scripts and drafting/editing (OpenAI, 2025). Plots use retained observations, and examples are actual saved carriers. Manuscript preparation adds no experiment or substitute imagery. The authors re-

Table S6: All five development payload/method combinations. Latencies are means of charged fresh-process encode/decode times. Fixed has three timing repetitions per payload; gating and A1 have one comparison on the 32-byte payload. Speedups are ratios of matched mean pair times. Per-row throughput is the accepted mean of individual source-bytes/pair-second ratios; the main headline instead pools bytes and times over nine fixed comparisons. All 11 comparisons and 22 recoveries preserve exact stream, carrier and source-byte equality. Repetitions do not increase the independent payload count.

Payload / method	Bytes / repeats	Reference encode / decode (s)	Graph encode / decode (s)	Pair speedup	Recovered B/s reference $\rightarrow$ graph	Equivalence
T1 / Fixed	32 / 3	88.97 / 89.28	17.38 / 17.28	5.14 $\times$	0.180 $\rightarrow$ 0.923	Exact
T1 / Gated	32 / 1	90.17 / 90.07	17.38 / 17.23	5.21 $\times$	0.178 $\rightarrow$ 0.925	Exact
T1 / A1	32 / 1	91.83 / 91.03	17.83 / 17.28	5.21 $\times$	0.175 $\rightarrow$ 0.911	Exact
T4 / Fixed	96 / 3	89.12 / 89.09	17.34 / 17.29	5.15 $\times$	0.539 $\rightarrow$ 2.772	Exact
T5 / Fixed	128 / 3	88.76 / 88.24	17.36 / 17.27	5.11 $\times$	0.723 $\rightarrow$ 3.696	Exact

main responsible for verification and submission decisions.

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